1. Decision, scope and revised findings

Evaluate whether adding live video to an existing neonatal specialist telephone pathway is worthwhile from organisational, health-system and societal perspectives. Existing data and explicit reasoning-based estimates are sufficient for a preliminary model. Neither prospective recruitment nor structured clinical elicitation is required. The source project and results snapshot, rather than the earlier v1 narrative, are authoritative for v2.

The unit of analysis is a consultation after birth, not every birth and not every retrieval. Both comparator arms retain specialist advice. The primary calculation estimates programme and pathway resource differences; it does not assume equivalent outcomes, a mortality effect, or automatic funding/cash recovery. Positive net saving means the video pathway uses fewer valued resources.

Measure	Pilot: 3 sites	Illustrative state: 33 spokes
Historical 2022 births	486	28,276
Forecast consultations/year	37.5	1,114.5
Deterministic programme cost	$14,052	$327,222
Deterministic net resource saving	$4,167	$137,912
PSA mean net saving	$2,428	$135,429
PSA median net saving	–$2,194	$9,692
PSA 95% interval	–$41,148 to +$65,642	–$1,011,529 to +$1,808,652
Probability cost saving	46.0%	50.7%

These probabilities are conditional on researcher-specified priors. They are not empirical posterior probabilities that the pilot works. No statistical test has established local transfer reduction. The change from the earlier 75% figure reflects a more conservative cost boundary, rescue/reliability effects, critical-case separation and explicit shared uncertainty—not newly observed clinical outcomes. [M1]

Organisational analysis separates who pays from whose resources are used. A CHHHS saving need not be a Queensland system saving, and released capacity need not reduce expenditure. Funding caps, block allocations and contract commitments require their own budget ledger. Costs accruing through extra survival can make a clinically valuable service more expensive.

2. Population and geographical extrapolation

Queensland Perinatal Data Collection table 8.09 (2022) provides 72 public-facility rows and 46,711 babies. Co-located birth centres are merged with their hospitals, preserving 65 sites and the total. Private hospitals have only an aggregate 13,741 births in this table; home births (350) and births before arrival (602) are not automatically device sites. The overall 61,404 is babies, not the 60,477 mothers. [QH2022]

Pilot site	2022 babies	Forecast consultations	Implied consultations / birth
Atherton	157	10–15	6.4–9.6%
Mareeba	134	10–15	7.5–11.2%
Innisfail	195	10–15	5.1–7.7%

The implied ratios combine historical births with a current service forecast. They are not observed neonatal illness or transfer rates. The 4–5 Atherton episodes have no supplied observation duration or complete denominator and therefore do not update an incidence parameter.

The primary state portfolio selects 33 historical non-hub public sites with at least 20 births (28,276 babies). Four historical ANZNN NICU hubs and 28 very-low-volume readiness locations are separate. All-public, rural-only, hub-capacity and reopened-Weipa scenarios are supplied. The 20-birth threshold is a modelling decision, not a policy or capability standard. [QH2022; ANZNN2023; WEIPA2026]

The 2023 ANZNN directory identifies four Queensland NICU members and seven SCN members. It is not a complete current service census. Its high-risk registration criteria select preterm, low-birthweight, ventilated, surgical or encephalopathic infants. These selected cases cannot be treated as the casemix of all rural consultations. No de-identified network graph has been reverse-mapped to a hospital.

Projected calls use the pilot midpoint of 37.5/486 historical births, multiplied by site births, an assumed capability factor (SCN 0.5; other large regional 0.65; rural/readiness 1), remote factor 1.25 and uptake 80%. Pilot forecasts are not scaled a second time. Birth-growth scenarios ±20%, uptake 40–100%, rural-only and critical-heavy cases expose dependence on these choices. The approach does not infer a causal incidence rate from the pilot.

Geographical transport priors are analyst assumptions: metro 85/10/5%, regional 60/25/15%, remote 15/35/50% (road/rotary/fixed). No actual route, dispatch or flight-time database is claimed. Newer 2025 birth-registration data were located but their content was not retrievable; registration transactions would also require reconciliation to perinatal birth definitions. The site CSV is therefore transparently historical.

3. Literature-to-parameter mapping

Primary source	Finding used	Model role / limit
Armfield 2012 [ARM2012]	5 avoided transfers / 19 consultations; 4 helicopter; $54,400 historical net saving	Queensland comparator and external benchmark, not a reliable local causal rate or current cost.
Haynes 2020 [HAY2020]	317 babies; video 64% vs phone 82%; adjusted OR 0.48 (0.26–0.90)	Log-OR uncertainty informs an attenuated, transportability-widened prior. Observational, term/late-preterm cohort.
Haynes 2021 [HAY2021]	Adjusted stabilisation difference –1.8 min (–16.0 to 12.4)	No proven time saving is monetised. Video cohort was sicker.
Yoo 2022 [YOO2022]	Decision model, 1,000 Monte Carlo runs; ROI 2.23; 75% cost saving	Method and state-scale comparator. 2017 US costs not transplanted to Queensland.
Thao 2022 [THAO2022]	Decision tree: US $16,878 vs $28,047 per patient	Transfer/transport/hospital structure; not independent clinical-effect evidence.
Lietz 2026 [LIETZ2026]	93 simulation participants; specialist support improved score; no phone/video/AR difference	Supports null incremental-video and usability scenarios. Simulation is not clinical equivalence proof.
Sandhu 2026 [SANDHU2026]	20 workshop participants; confidence increased	Training/readiness costs and implementation, not mortality or transfer-effect estimation.

Searches were targeted PubMed discovery plus exact-identifier verification and reference-led public-source searches to 27 October 2026. This is not an exhaustive systematic review. These paper extractions were verified from abstracts; full-text model tables and original price years remain incompletely extracted. The evidence CSV records eligibility, use, transportability and limitations. Two economic models using the same clinical effect are not counted as two independent effect estimates.

Haynes log(OR) = log(0.48); SE = [log(0.90) – log(0.26)] / 3.92. The reference model retains half the log effect, giving OR 0.693, and adds an assumed transportability SD of 0.35. This is a transparent analyst prior, not a fitted meta-analysis. Published-effect, null and zero-centred sceptical structures are evaluated separately. The comparative figure labels the study confidence interval differently from the model prior interval.

NeoSTRESS [NEOSTRESS2020] measured sound on 45 outbound, non-patient retrieval legs: 25 road, 9 rotary and 11 fixed wing. It is a convenience environmental study, not a Queensland or FNQ service census. Its mode proportions are retained only as an external comparison, not a high-precision mode prevalence estimate.

4. Retrieval costs, indexation and dual pricing

A 2024 primary study of Torres Strait skin/soft-tissue infection care in 2018–2019 reports Queensland Health aeromedical cost $18,670. This all-age, non-neonatal mixed-air estimate is a geographically relevant anchor, not an RSQ contract tariff or marginal saving. The underlying tariff year is not separately explicit: 2019 is assumed, with 2018 tested. [AIR2019]

Indexation component	Value / role
2019 annual CPI, old base	115.125
September 2025 quarter, old base	143.6
Same quarter, new base	99.73
June 2026 monthly CPI, new base	102.03
Formula	18,670 × 102.03 / [115.125 × (99.73 / 143.6)]
June 2026 price-point anchor	$23,824.91

The reference-period bridge prevents dividing incompatible CPI bases. All-groups CPI is a purchasing-power approximation, not a neonatal transport price index. The rounded bridge and annual-to-month reference choice are explicit. An additional 30% air-cost scenario addresses labour/fuel/contract divergence. June 2026 is the selected price point, not a claim to October prices. [CPI2019; CPIBASE; CPI2026]

Rotary 1.2× and fixed-wing 0.9× the air anchor are assumptions. Road cost $4,000 is an illustrative resource build: 2 staff × 10 hours × $150, plus 400 km × $0.90, $400 coordination and $240 equipment. Its $2,000–$8,000 range is not a published QAS tariff. Only 50% of allocated transport resource cost is assumed avoidable (Beta(2,2) in PSA; 25–100% deterministic scenarios). Net hospital difference is $2,000, with a range allowing extra local expenditure.

NEP26 is $7,418 per NWAU(26), with AR-DRG v12.0. Queensland uses QWAU and QEP, with local adjustments and GEN_WAU calculations. Keep resource cost, NWAU × NEP, QWAU × QEP and actual funding flows as four distinct ledgers. Do not add the two funding benchmarks together or treat a funding reduction as an expenditure saving. QEP and local QWAU inputs remain missing unless verified; the 0.9/1.0/1.1 ratio grid is sensitivity analysis only. [NEP26; QHFUND26; QAO2025]

The user’s mchs repository is the appropriate NWAU tool. Its 2026 documentation explicitly states that NEP26 calculator parity is not yet validated. This version therefore implements a year/version guard and an interface specification; it does not execute mchs or claim validated 2026 patient-level outputs. Use verified GEN_WAU/official calculator exports for full episode valuation. [MCHS]

5. Uncertainty, structure and clinical consequences

The core is a stratified expected-value model, not a discrete-event simulator. Low/moderate/critical shares are assumed 35/50/15%; baseline transfer probabilities 30/65/98%. Critical cases receive no transfer-avoidance credit. Failed video falls back to telephone. Five percent of apparently avoided transfers require later rescue, incurring an additional penalty. Acute mortality is not inferred from procedure confidence or reduced transfers.

For each of pilot and state portfolios, 20,000 fixed-seed draws jointly vary effect, casemix, baseline transfer, reliability, rescue, resource costs, programme cost and activity. Common draws retain correlation across sites: an uncertain clinical effect does not become known simply because a state has more births. Separate predictive simulations add annual consultation/transfer count variation without presenting it as new evidence. Distributions, source IDs and rationale are in parameter_register.csv. [M1]

The 58 deterministic comparisons vary no/published/increased-transfer effects; critical-heavy and critical-only cases; no/negative hospital offset; 25/50/100% avoidable transport fractions; road/air-heavy geography; 10/15 consultations per pilot site; uptake; birth counts; connection reliability; delayed rescue; tablet/current/cart capital; sunk capital; shared hub overhead and capacity; useful life and discounting; historical price year and extra inflation. All-public, rural-only and Weipa reopening portfolios additionally test population boundaries.

Further structural work remains warranted: empirically supported origin–destination routes and flight/crew times; antenatal maternal transfer and unplanned outborn substitution; repeated consultations per infant; common-mode network outages; response delays at specialist hubs; cohort-specific differential effects; longer-term morbidity conditional on severity; distributional equity and caregiver spillover; and alternative correlation/copula assumptions. These are recorded as open tracks, not silently claimed implemented.

The exploratory 108-point survival frontier permits additional survivors with major morbidity and higher costs. It varies absolute survival gain (0, 0.1%, 0.5%, 1%), major-morbidity share (10%, 35%, 70%), discount (0%, 3%, 5%) and illustrative willingness-to-pay ($0/$50,000/$100,000). Forty-year constant survival, utilities 0.9/0.5 and annual costs $1,000/$20,000 are assumptions. No lifetime incremental cost-effectiveness claim is made. Utility numbers do not define the worth of a disabled person.

Report net monetary benefit only for the explicitly valued scenario. Cost-only EVPI estimates the expected resource consequence of the invest/not-invest decision under these priors; it excludes unquantified clinical value and is not automatically a research budget. Rank correlations identify sensitivity, not causal effects or EVPPI. [M1]

6. Reproducibility, assurance and remaining gates

The restored five-test v1 was not backed by a CI workflow or dependency lock. V2 adds 75 passing tests and 96% branch-inclusive coverage of the new model, a data/provenance/reproducibility harness, fixed seeds, source and input hashes, a version gate for mchs, and Conductor context, specification, plan, decisions and verification history. The review was by the implementing author, not an independent reviewer.

Observed failures were addressed: a zero-offset test incorrectly omitted the still-active rescue penalty; its fixture was corrected. PSA initially ignored custom critical-only casemix; the model was corrected and the regression test now passes. Coverage execution was changed to the coverage CLI to avoid a hidden pytest-cov dependency in the online workflow.

Current official GitHub Actions were checked and pinned by commit SHA: checkout 7.0.1, setup-python 7.1.0 and upload-artifact 7.0.2. The workflow has read-only default permissions, no persisted checkout credentials, job timeouts/concurrency, Python 3.13/3.14/3.15 matrix, a dependency-resolution/hash stage, lint, coverage, generative properties, vulnerability audit/SBOM and an approval-gated release. No clinical data or corporate templates are automatically published.

Online target versions are separately recorded from the tested environment. Targets include NumPy 2.5.4, pytest 9.1.1, coverage 7.17.0, Ruff 0.17.3, Hypothesis 6.170.12, pip-audit 2.10.1 and uv 0.13.5. These targets were not installed here. Local execution used Python 3.13.5, NumPy 2.3.5, pytest 9.0.2 and coverage 7.13.3. requirements-latest.in is a target specification, not a successfully resolved transitive hash lock.

External DNS failures prevented dependency resolution and actual Conductor submodule checkout. Ruff, Hypothesis, ty and pip-audit were unavailable locally. Hosted Actions, new-interpreter matrices, vulnerability checks and release approval were not run. The verification report marks these BLOCKED; no all-CI-green, latest-environment or production-ready claim is made. No remote repository was created or changed.

The configured Conductor submodule remains pinned to edithatogo/conductor-next at 137c1e41ff9c1dc75ff277b9582460a0bd8c4768, but its working tree is unpopulated. Standard conductor files are committed locally. The bundle preserves prior history; the ZIP includes the updated source/evidence/outputs. A connected execution must populate the submodule, resolve and review the lock, run every configured gate and obtain a real hosted receipt before release.

Workbook inputs and the deterministic pilot calculation are editable formulas. State scenario tables and PSA are verified exported snapshots: editing a workbook cell does not rerun Python simulations. Use python scripts/run_v2.py and regenerate artifacts to update those outputs. National transferability is a modelling framework, not a completed Australia-wide site model.

7. Source register and exact provenance

Identifiers below match evidence/sources.csv and the parameter map. URLs identify primary or official sources. Access/check date: 27 October 2026. Source content, publication dates and price years remain distinct. Local information is not a public publication.

[QH2022] Queensland Health. Queensland Perinatal Statistics 2022, Chapter 8. Source period: 2022.
https://www.health.qld.gov.au/__data/assets/pdf_file/0020/1378010/8-fac2022x.pdf
Limit: Table 8.09 screenshot verified; all 72 public rows transcribed; private aggregate only; historical not 2026

[BDM2025] Queensland Registrar. Births by hospital 2025. Source period: 2025.
https://www.data.qld.gov.au/dataset/births-by-hospital/resource/51505dd2-c23d-4369-8ad8-c1d501c24e88
Limit: Metadata verified; CSV/API access failed; no counts imported; registration transactions differ from perinatal births

[ANZNN2023] UNSW/ANZNN. ANZNN report 2023. Source period: 2023.
https://www.unsw.edu.au/content/dam/images/medicine-health/npesu/research/annual-reports/2026-03-report-of-the-australian-and-new-zealand-neonatal-network-2023-excl-follow-up.pdf
Limit: Published March 2026; 4 QLD NICUs and 7 participating SCNs located; selected high-risk cohort, not complete birth denominator

[AIR2019] RRH primary research. SSTI healthcare resource use, Torres Strait. Source period: 2018-2019.
https://www.rrh.org.au/journal/article/8572/
Limit: QH-provided aeromedical cost $18670; audit period 2018-2019, tariff price year not separately explicit; 2019 price base assumed, 2018 sensitivity

[CPI2019] ABS. CPI June-quarter 2025 historical series. Source period: 2019.
https://www.abs.gov.au/statistics/economy/price-indexes-and-inflation/consumer-price-index-australia/jun-quarter-2025
Limit: 2019 quarterly indices 114.1,114.8,115.4,116.2; annual average115.125; 2018 annual average113.3; old base2011-12

[CPIBASE] ABS. Re-referencing the quarterly CPI. Source period: 2026.
https://www.abs.gov.au/statistics/detailed-methodology-information/information-papers/re-referencing-quarterly-consumer-price-index
Limit: Old Sep-quarter2025=143.6; new=99.73; conversion99.73/143.6. Approximate published rounded indices.

[CPI2026] ABS. CPI June 2026. Source period: 2026.
https://www.abs.gov.au/statistics/economy/price-indexes-and-inflation/consumer-price-index-australia/jun-2026
Limit: Monthly June2026 index102.03 on Sep-month2025=100. Explicit June2026 price point, not October2026 latest prices.

[QHFUND26] Queensland Health. HHS Funding and Purchasing Guidelines. Source period: 2026.
https://www.publications.qld.gov.au/dataset/e1c2648f-eb8e-4e7f-a0d7-42604cd9212f/resource/c8b19f98-6a34-4db7-a660-1e60391e4325/download/hhs-funding-and-purchasing-guidelines.pdf
Limit: GEN_WAU calculates NWAU/QWAU; QEP local funding; no verified current numeric QEP; detailed model internal

[NEP26] IHACPA. National Efficient Price Determination 2026-27. Source period: 2026-27.
https://www.ihacpa.gov.au/resources/national-efficient-price-determination-2026-27
Limit: NEP7418, AR-DRG12.0; national benchmark not marginal savings

[MCHS] Dylan Mordaunt GitHub. mchs README and NWAU interface. Source period: 2026.
https://github.com/edithatogo/mchs
Limit: Head9fc8d0f33df2aabaef8d4a10f25df12d41ff033e; README validates2024/2025 only; 2026 and QWAU not certified; optional interface rather than dependency

[ELC2026] CHHHS. ELC item2.1.7 Panda Vision. Source period: 2026-10-21.
internal://CHHHS/ELC/2026-10-21/item2.1.7
Limit: Installed three-site system; $10000 indicative; full confidential pack excluded from new public-data exports

[USER2026] Dylan Mordaunt. User clarification. Source period: 2026-10-27.
conversation://panda-vision/2026-10-27
Limit: 4-5 Atherton cases; 10-15 anticipated consultations per site; no structured local elicitation prerequisite

[NEOSTRESS2020] Gilmour et al.. NeoSTRESS neonatal retrieval sound study. Source period: 2020.
https://pubmed.ncbi.nlm.nih.gov/32949204/
Limit: Brisbane45outbound non-patient legs:25road,11fixed,9rotary; convenience sample NOT dispatch census or FNQ mode probabilities

[QAO2025] Queensland Audit Office. Health 2025, tabled 20 January 2026. Source period: 2024-25.
https://www.qao.qld.gov.au/reports-resources/reports-parliament/health-2025
Limit: Funding price and actual costs differ; not a verified numeric QEP input

[WEIPA2026] Torres and Cape HHS. Weipa maternity service celebrates 50th baby birth. Source period: 2025.
https://www.torres-cape.health.qld.gov.au/about-us/news/weipa-maternity-service-celebrates-50th-baby-birth
Limit: 32 births in2025; separate reopened-service scenario, not merged silently into2022 denominator

8. Primary publication references

[ARM2012] Armfield et al. 2012. PMID 23148301.
https://pubmed.ncbi.nlm.nih.gov/23148301/
Design: Queensland observational economic study. Review depth: abstract verified.

[HAY2020] Haynes et al. 2020. PMID 32081766.
https://pubmed.ncbi.nlm.nih.gov/32081766/
Design: Observational comparison. Review depth: abstract verified.

[HAY2021] Haynes et al. 2021. PMID 33691080.
https://pubmed.ncbi.nlm.nih.gov/33691080/
Design: Observational stabilisation study. Review depth: abstract verified.

[YOO2022] Yoo et al. 2022. PMID 35074308.
https://pubmed.ncbi.nlm.nih.gov/35074308/
Design: Decision model/PSA/state extrapolation. Review depth: abstract verified.

[THAO2022] Thao et al. 2022. PMID 35235430.
https://pubmed.ncbi.nlm.nih.gov/35235430/
Design: Decision tree/thresholds. Review depth: abstract verified.

[SANDHU2026] Sandhu et al. 2026. PMID 42313572.
https://pubmed.ncbi.nlm.nih.gov/42313572/
Design: Training/workshop. Review depth: abstract verified.

[LIETZ2026] Lietz et al. 2026. PMID 42647428.
https://pubmed.ncbi.nlm.nih.gov/42647428/
Design: Randomised crossover simulation. Review depth: abstract verified.

[M1] Panda Vision v2 analysis: outputs/v2/results_v2.json; source, parameter and scenario CSVs; fixed seed 20261027. Reproducible source and verification receipts accompany the project.